Module 2

For fraction = 0.05

```
Using MSE threshold = 0.00047 (5% of true value range, not the

--- MC ---
  B=   50  MSE=0.02108  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02102  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01894  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01463  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01338  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01097  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.00313  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.00254  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.00352  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.00220  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.00263  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.00183  steps_used=14002  episodes_used=1824.6
  B_min (steps) = None | episodes consumed at B_min = None

--- TD(0) ---
  B=   50  MSE=0.02242  steps_used=50  episodes_used=6.8
  B=   75  MSE=0.02242  steps_used=75  episodes_used=9.4
  B=  125  MSE=0.02103  steps_used=125  episodes_used=16.6
  B=  250  MSE=0.01723  steps_used=250  episodes_used=34.6
  B=  500  MSE=0.01636  steps_used=500  episodes_used=68.6
  B= 1000  MSE=0.01376  steps_used=1000  episodes_used=135.8
  B= 2000  MSE=0.00645  steps_used=2000  episodes_used=264.2
  B= 4000  MSE=0.00291  steps_used=4000  episodes_used=522.8
  B= 8000  MSE=0.00214  steps_used=8000  episodes_used=1050.0
  B=10000  MSE=0.00106  steps_used=10000  episodes_used=1314.0
  B=12000  MSE=0.00081  steps_used=12000  episodes_used=1568.6
  B=14000  MSE=0.00090  steps_used=14000  episodes_used=1824.2
  B_min (steps) = None | episodes consumed at B_min = None

--- n-step TD (n=4) ---
  B=   50  MSE=0.02101  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02101  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01961  steps_used=132  episodes_used=17.4
```

```
 B=  250  MSE=0.01665  steps_used=254  episodes_used=35.0
 B=  500  MSE=0.01606  steps_used=505  episodes_used=69.4
 B= 1000  MSE=0.01493  steps_used=1006  episodes_used=136.0
 B= 2000  MSE=0.01010  steps_used=2003  episodes_used=264.8
 B= 4000  MSE=0.01116  steps_used=4005  episodes_used=523.2
 B= 8000  MSE=0.01224  steps_used=8004  episodes_used=1050.8
 B=10000  MSE=0.01299  steps_used=10005  episodes_used=1314.2
 B=12000  MSE=0.01591  steps_used=12004  episodes_used=1568.8
 B=14000  MSE=0.01326  steps_used=14002  episodes_used=1824.6
 B_min (steps) = None | episodes consumed at B_min = None
```

For fraction = 0.1

```
Using MSE threshold = 0.00188 (5% of true value range, not the raw

--- MC ---
  B=   50  MSE=0.02108  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02102  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01894  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01463  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01338  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01097  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.00313  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.00254  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.00352  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.00220  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.00263  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.00183  steps_used=14002  episodes_used=1824.6
  B_min (steps) = 14000 | episodes consumed at B_min = 1824.6

--- TD(0) ---
  B=   50  MSE=0.02242  steps_used=50  episodes_used=6.8
  B=   75  MSE=0.02242  steps_used=75  episodes_used=9.4
  B=  125  MSE=0.02103  steps_used=125  episodes_used=16.6
  B=  250  MSE=0.01723  steps_used=250  episodes_used=34.6
  B=  500  MSE=0.01636  steps_used=500  episodes_used=68.6
  B= 1000  MSE=0.01376  steps_used=1000  episodes_used=135.8
  B= 2000  MSE=0.00645  steps_used=2000  episodes_used=264.2
  B= 4000  MSE=0.00291  steps_used=4000  episodes_used=522.8
  B= 8000  MSE=0.00214  steps_used=8000  episodes_used=1050.0
  B=10000  MSE=0.00106  steps_used=10000  episodes_used=1314.0
  B=12000  MSE=0.00081  steps_used=12000  episodes_used=1568.6
  B=14000  MSE=0.00090  steps_used=14000  episodes_used=1824.2
  B_min (steps) = 10000 | episodes consumed at B_min = 1314.0

--- n-step TD (n=4) ---
  B=   50  MSE=0.02101  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02101  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01961  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01665  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01606  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01493  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.01010  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.01116  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.01224  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.01299  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.01591  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.01326  steps_used=14002  episodes_used=1824.6
  B_min (steps) = None | episodes consumed at B_min = None
```

For fraction = 0.15

```
Using MSE threshold = 0.00423 (5% of true value range, not the r

--- MC ---
  B=   50  MSE=0.02108  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02102  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01894  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01463  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01338  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01097  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.00313  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.00254  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.00352  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.00220  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.00263  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.00183  steps_used=14002  episodes_used=1824.6
  B_min (steps) = 2000 | episodes consumed at B_min = 264.8

--- TD(0) ---
  B=   50  MSE=0.02242  steps_used=50  episodes_used=6.8
  B=   75  MSE=0.02242  steps_used=75  episodes_used=9.4
  B=  125  MSE=0.02103  steps_used=125  episodes_used=16.6
  B=  250  MSE=0.01723  steps_used=250  episodes_used=34.6
  B=  500  MSE=0.01636  steps_used=500  episodes_used=68.6
  B= 1000  MSE=0.01376  steps_used=1000  episodes_used=135.8
  B= 2000  MSE=0.00645  steps_used=2000  episodes_used=264.2
  B= 4000  MSE=0.00291  steps_used=4000  episodes_used=522.8
  B= 8000  MSE=0.00214  steps_used=8000  episodes_used=1050.0
  B=10000  MSE=0.00106  steps_used=10000  episodes_used=1314.0
  B=12000  MSE=0.00081  steps_used=12000  episodes_used=1568.6
  B=14000  MSE=0.00090  steps_used=14000  episodes_used=1824.2
  B_min (steps) = 4000 | episodes consumed at B_min = 522.8

--- n-step TD (n=4) ---
  B=   50  MSE=0.02101  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02101  steps_used=77  episodes_used=9.8
```

```
B=  125  MSE=0.01961  steps_used=132  episodes_used=17.4
B=  250  MSE=0.01665  steps_used=254  episodes_used=35.0
B=  500  MSE=0.01606  steps_used=505  episodes_used=69.4
B= 1000  MSE=0.01493  steps_used=1006  episodes_used=136.0
B= 2000  MSE=0.01010  steps_used=2003  episodes_used=264.8
B= 4000  MSE=0.01116  steps_used=4005  episodes_used=523.2
B= 8000  MSE=0.01224  steps_used=8004  episodes_used=1050.8
B=10000  MSE=0.01299  steps_used=10005  episodes_used=1314.2
B=12000  MSE=0.01591  steps_used=12004  episodes_used=1568.8
B=14000  MSE=0.01326  steps_used=14002  episodes_used=1824.6
B_min (steps) = None | episodes consumed at B_min = None
```

For fraction = 0.20

```
Using MSE threshold = 0.00752 (5% of true value range, not the r

--- MC ---
  B=   50  MSE=0.02108  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02102  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01894  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01463  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01338  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01097  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.00313  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.00254  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.00352  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.00220  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.00263  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.00183  steps_used=14002  episodes_used=1824.6
  B_min (steps) = 2000 | episodes consumed at B_min = 264.8

--- TD(0) ---
  B=   50  MSE=0.02242  steps_used=50  episodes_used=6.8
  B=   75  MSE=0.02242  steps_used=75  episodes_used=9.4
  B=  125  MSE=0.02103  steps_used=125  episodes_used=16.6
  B=  250  MSE=0.01723  steps_used=250  episodes_used=34.6
  B=  500  MSE=0.01636  steps_used=500  episodes_used=68.6
  B= 1000  MSE=0.01376  steps_used=1000  episodes_used=135.8
  B= 2000  MSE=0.00645  steps_used=2000  episodes_used=264.2
  B= 4000  MSE=0.00291  steps_used=4000  episodes_used=522.8
  B= 8000  MSE=0.00214  steps_used=8000  episodes_used=1050.0
  B=10000  MSE=0.00106  steps_used=10000  episodes_used=1314.0
  B=12000  MSE=0.00081  steps_used=12000  episodes_used=1568.6
  B=14000  MSE=0.00090  steps_used=14000  episodes_used=1824.2
  B_min (steps) = 2000 | episodes consumed at B_min = 264.2

--- n-step TD (n=4) ---
  B=   50  MSE=0.02101  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02101  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01961  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01665  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01606  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01493  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.01010  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.01116  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.01224  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.01299  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.01591  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.01326  steps_used=14002  episodes_used=1824.6
  B_min (steps) = None | episodes consumed at B_min = None
```

For fraction = 0.25

```
Using MSE threshold = 0.01175 (5% of true value range, not the r

--- MC ---
  B=   50  MSE=0.02108  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02102  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01894  steps_used=132  episodes_used=17.4
  B=  250  MSE=0.01463  steps_used=254  episodes_used=35.0
  B=  500  MSE=0.01338  steps_used=505  episodes_used=69.4
  B= 1000  MSE=0.01097  steps_used=1006  episodes_used=136.0
  B= 2000  MSE=0.00313  steps_used=2003  episodes_used=264.8
  B= 4000  MSE=0.00254  steps_used=4005  episodes_used=523.2
  B= 8000  MSE=0.00352  steps_used=8004  episodes_used=1050.8
  B=10000  MSE=0.00220  steps_used=10005  episodes_used=1314.2
  B=12000  MSE=0.00263  steps_used=12004  episodes_used=1568.8
  B=14000  MSE=0.00183  steps_used=14002  episodes_used=1824.6
  B_min (steps) = 1000 | episodes consumed at B_min = 136.0

--- TD(0) ---
  B=   50  MSE=0.02242  steps_used=50  episodes_used=6.8
  B=   75  MSE=0.02242  steps_used=75  episodes_used=9.4
  B=  125  MSE=0.02103  steps_used=125  episodes_used=16.6
  B=  250  MSE=0.01723  steps_used=250  episodes_used=34.6
  B=  500  MSE=0.01636  steps_used=500  episodes_used=68.6
  B= 1000  MSE=0.01376  steps_used=1000  episodes_used=135.8
  B= 2000  MSE=0.00645  steps_used=2000  episodes_used=264.2
  B= 4000  MSE=0.00291  steps_used=4000  episodes_used=522.8
  B= 8000  MSE=0.00214  steps_used=8000  episodes_used=1050.0
  B=10000  MSE=0.00106  steps_used=10000  episodes_used=1314.0
  B=12000  MSE=0.00081  steps_used=12000  episodes_used=1568.6
  B=14000  MSE=0.00090  steps_used=14000  episodes_used=1824.2
  B_min (steps) = 2000 | episodes consumed at B_min = 264.2

--- n-step TD (n=4) ---
  B=   50  MSE=0.02101  steps_used=55  episodes_used=7.2
  B=   75  MSE=0.02101  steps_used=77  episodes_used=9.8
  B=  125  MSE=0.01961  steps_used=132  episodes_used=17.4
```

```
B=  250  MSE=0.01665  steps_used=254  episodes_used=35.0
B=  500  MSE=0.01606  steps_used=505  episodes_used=69.4
B= 1000  MSE=0.01493  steps_used=1006  episodes_used=136.0
B= 2000  MSE=0.01010  steps_used=2003  episodes_used=264.8
B= 4000  MSE=0.01116  steps_used=4005  episodes_used=523.2
B= 8000  MSE=0.01224  steps_used=8004  episodes_used=1050.8
B=10000  MSE=0.01299  steps_used=10005  episodes_used=1314.2
B=12000  MSE=0.01591  steps_used=12004  episodes_used=1568.8
B=14000  MSE=0.01326  steps_used=14002  episodes_used=1824.6
B_min (steps) = 2000 | episodes consumed at B_min = 264.8
```

Algorithm (for fraction = 0.25) (threshold = 0.01175)	B_{min} (steps)	Mean episodes consumed at B_{min}	Mean MSE at B=500
TD(0)	2000	264.2	0.01636
n-step TD (n=4)	2000	264.8	0.01606
Monte Carlo	1000	136.0	0.01338

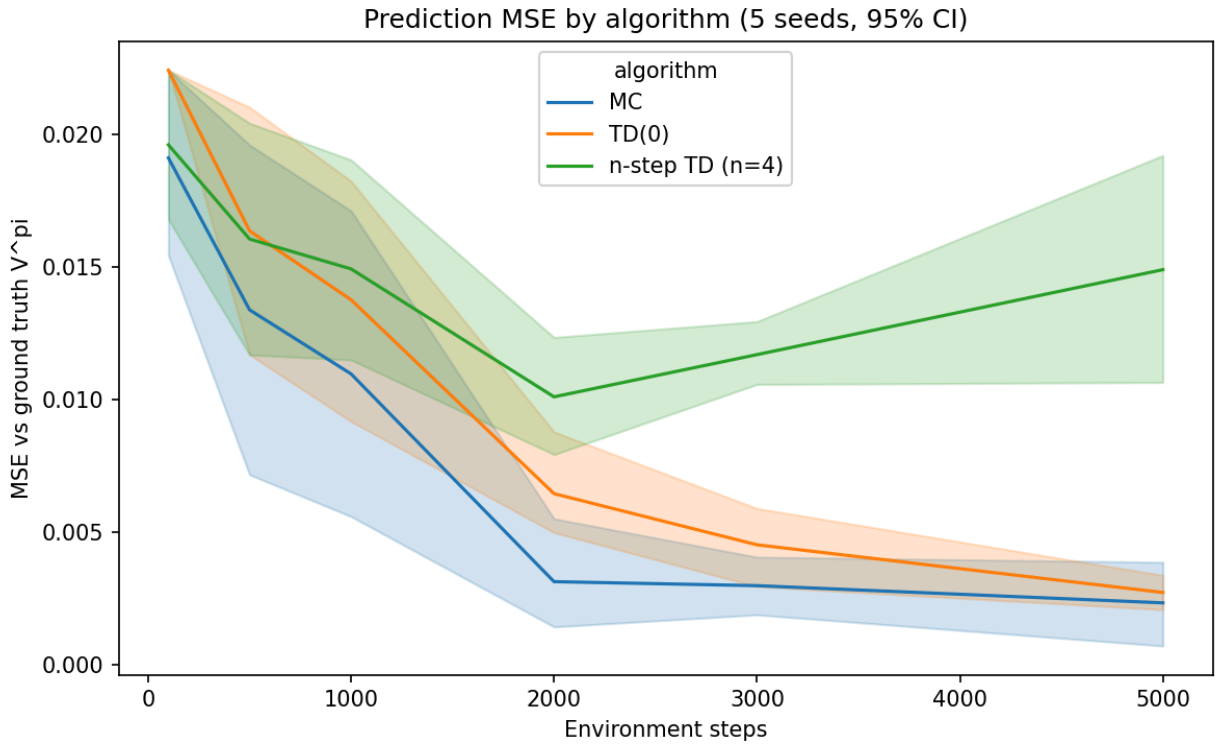
Module 3

```
=== Horizon sweep (n-step TD) ===
  deterministic:
    n=  1  MSE=0.00218
    n=  2  MSE=0.00626
    n=  4  MSE=0.01086
    n=  8  MSE=0.01279
    n= 16  MSE=0.01289
    n= 32  MSE=0.01289
  stochastic:
    n=  1  MSE=0.00254
    n=  2  MSE=0.00913
    n=  4  MSE=0.01490
    n=  8  MSE=0.01815
    n= 16  MSE=0.01820
    n= 32  MSE=0.01825

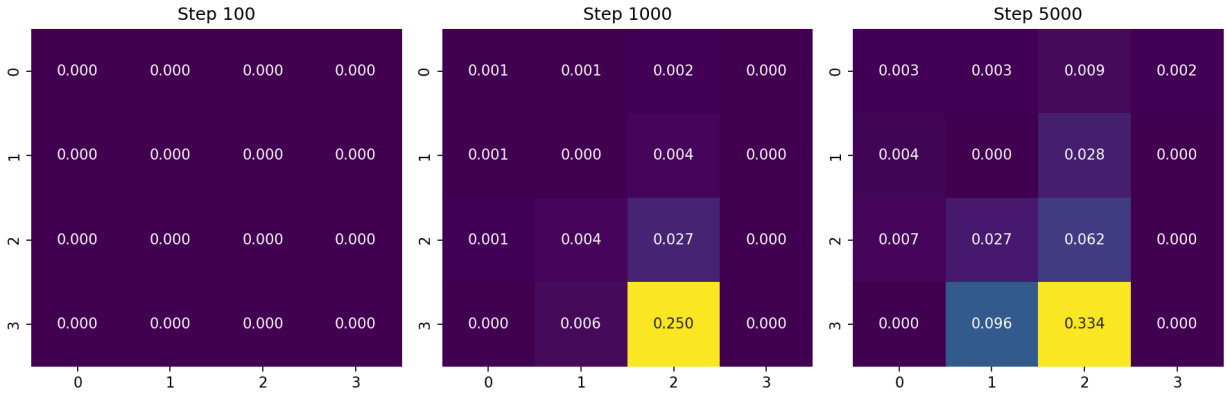
=== LR schedule comparison ===
  MC:
    constant              MSE=0.00233
    linear_decay          MSE=0.00070
    harmonic_decay        MSE=0.00129
    state_visit_count     MSE=0.00382
  TD(0):
    constant              MSE=0.00272
    linear_decay          MSE=0.00134
    harmonic_decay        MSE=0.00191
    state_visit_count     MSE=0.04416
  n-step TD (n=4):
    constant              MSE=0.01490
    linear_decay          MSE=0.01315
    harmonic_decay        MSE=0.01447
    state_visit_count     MSE=0.01866
```

Module 4

Multi-seed performance curves



Spatial value heatmaps



Module 4

1. The bootstrapping trap of alpha = 1/N(s). Explain why alpha = 1/N(s) computes exact sample averages for Monte Carlo, but can cause TD(0) to get stuck if early, noisy visits bake a high-bias bootstrapped estimate into V(s) before enough data has accumulated to correct it.

Ans. Alpha = 1/N(s) computes exact sample averages for Monte Carlo but can cause TD(0) to get stuck if early, noisy visits bake a high-bias bootstrapped estimate into V(s) before enough data has accumulated to correct it because if you look at Module 3, the MC algorithm has a very small MSE (0.00382), which is very close to the alpha value but the TD(0) algorithm has a very large MSE in comparison (0.04416) is much further. TD(0) suffers from being very vulnerable to bias due to early training data being biased and getting stuck making sub-optimal choices, while the MC algorithm is relatively unbiased since it has a high variance instead (bias-variance tradeoff).

2. State-wise vs. global decay. Why does alpha = 1/N(s) allow rarely-visited states near the goal to update rapidly late in training, whereas a global schedule alpha_t artificially throttles their learning rate?

Ans. As can be seen in the multi-seed performance curves, the n-step TD algorithm or globe decay does not perform well at extended step rates, since early training data biases it to certain early states and biases it against later states near the gloab but state-wise decay algorithms don't have this bias and allows the learning rate to be consistent and allow the later, rarely-visited stated to update late in training. The module 3 results also reflect this.

3. Stochastic horizon shift. Why does the optimal step horizon n shrink when moving from a deterministic environment to a stochastic one? Use your own horizon-sweep numbers from both settings. Note: V_pi itself is identical between the two settings for this policy (see Section 2) — your answer should be about variance in the data each n-step target is built from, not about the target values differing.

Ans. The optimal step horizon n shrinks when moving from a deterministic environment to a stochastic one to reduce variance in the algorithm caused by slippery tiles/ice squares as the n increases. So, the step horizon has to be shrunk to account for the lessen the vairance caused by the noise introduced by the slipper tiles.

4. Noise floor identification. Why does a constant learning rate alpha = 0.1 fail to converge to zero error in a stochastic environment, producing a constant-variance "noise floor"? This is a different source of variance from the seed-to-seed sampling noise in your Module 4 confidence-interval bands — don't conflate the two.

Ans. A constant learning rate alpha = 0.1 fails to converge to zero error in a stochastic environment since every action has an associated failure/slipping rate with it regardless. In such an environment, where slipping/failure is inbuilt into the scenario/problem, the learning rate will never converge to zero even if the constant learning rate being 0.1. This is reflected in the multi-seed performance curves where the MSE never reach zero and there is essentially a noise-floor present there.

5. Steps vs. episodes. Using your Module 2 table, compare the algorithms' ranking by step-based B_{min} against their ranking by mean episodes consumed at that budget. Do the rankings agree? If not, explain the discrepancy in terms of how MC vs. TD(0)/n-step TD make use of an episode's worth of data.

Ans. MC is first with B_{min} (steps) = 1000 steps and Mean episodes consumed at B_{min} = 136.0, then it is very close but TD(0) is second with B_{min} (steps) = 2000 steps and Mean episodes consumed at B_{min} = 264.2 followed by n-step TD (n=4) in third with B_{min} (steps) = 2000 steps and Mean episodes consumed at B_{min} = 264.8. The ranking for all of them by the step-based B_{min} [136.0, 264.2, 264.8] against their ranking by mean episodes consumed at that budget [1000, 2000, 2000] agree